

4 Appendix 1: AO-Cat Classes

4.1 AO_Entity

Subclass of:

Superclass of: AO_Resource, AO_Object, AO_Concept, AO_Event, AO_Agent,
AO_Spatial_Region, AO_Temporal_Region

Scope Note: AO_Entity has as instances all entities in the ARIADNE infrastructure

Domain of: has_identifier
has_type
has_title
has_description

Maps to: AO_Resource is a subclass of crm:E1_CRM_Entity

4.2 AO_Resource

Subclass of: AO_Entity

Superclass of: AO_Data_Resource; AO_Service

Scope Note: AO_Resource has as instances all archaeological resources that are represented in the AC. These resources are categorized in the following classes (see Figure below):

- AO_Data_Resource, representing all types of digital data object described in the Catalogue for discovery, access and integration. Two disjoint classes of data resources are further distinguished:
 - AO_Individual_Data_Resource, including all data resources that are considered as not further decomposed; these include datasets, databases, GIS, and so on.
 - AO_Collection, representing aggregations of data resources considered as wholes, possibly including other collections.
- AO_Service, representing the services owned by the A+ partners and, possibly, made available to other partners within the project on the ARIADNE infrastructure. This class is also suitable for describing new services created within the project.

Examples: AO_Data_Resource: a stratigraphic database, a collection of images from an excavation event, a report on the discovery of Roman burials in Provence
AO_Service: the ARIADNE Visual Media service.

Domain of: was_issued
was_modified
has_publisher
has_contributor
has_creator

has_owner
has_responsible
has_visual_component

Maps to: AO_Resource is a subclass of crm:E1_CRM_Entity

The four classes of resources mentioned above are described in detail in the rest of this Section.

4.2.1 AO_Data_Resource

Subclass of: AO_Resource

Superclass of: AO_Collection, AO_Individual_Data_Resource

Scope Note: This class specializes the AO_Resource class and is intended to describe the digital resources of the ARIADNE RI that are data. The type of an instance of this class (e.g., GIS, database, collection and the like) is specified through the has_type property. This is an abstract class: its instances are all and only the instances of its subclasses.

Examples: a stratigraphic database, a collection of images from an excavation event, a report on the discovery of Roman burials in Provence.

Maps to: AO_Data_Resource is a subclass of crmpe:PE22_Persistent_Dataset (which is a subclass of crm:E73_Information_Object) and is a subclass of crm:E33_Linguistic_Object

Domain of: has_original_id
is_about
has_ARIADNE_subject
has_native_subject
has_derived_subject
has_language
was_created_on
has_landing_page
has_access_policy
has_access_rights
has_extent
has_temporal_coverage
has_spatial_coverage

4.2.2 AO_Individual_Data_Resource

Subclass of: AO_Data_Resource

Superclass of: AO_Document

Scope Note: This class is a specialization of the class AO_Data_Resource, and has as instances data resources that from the ARIADNE RI point of view are atomic, that is no further decomposed. The complementary class is that of AO_Collection (see below), having as instances data resources that are composed of other data resources.

Examples: An archaeological record. A dataset. An image.

Maps to: AO_Individual_Data_Resource is implicitly a subclass of crmpe:PE22_Persistent_Dataset (being a subclass of AO_Data_Resource)

4.2.3 AO_Document

Subclass of: AO_Individual_Data_Resource

Superclass of:

Scope Note: This class is a specialization of the class AO_Individual_Data_Resource, and has as instances documents of interest to the ARIADNE RI.

Examples: A grey literature report. A scientific journal issue.

Maps to: AO_Document is implicitly a subclass of crmpe:PE22_Persistent_Dataset (being a subclass of AO_Individual_Data_Resource)

4.2.4 AO_Collection

Subclass of: AO_Data_Resource

Superclass of:

Scope Note: This class is a specialization of the class AO_Data_Resource, and has as instances collections in the archaeological and archaeological-related domains. An archaeological collection is an aggregation of resources, called the members in the collection, instances of AO_Data_Resource. As a consequence, collections can have other collections as members.

Examples: A group of resources documenting an excavation and composed of a report, a stratigraphic database and a collection of images.

Maps to: Collection is implicitly a subclass of crmpe:PE22_Persistent_Dataset (being a subclass of AO_Data_Resource)

Domain of: has_part

4.2.5 AO_Service

Subclass of: AO_Resource

Superclass of:

Scope Note: This class specializes the AO_Resource class and is intended to describe services in the archaeological domain. A service is “an offer by some actor of their willingness and ability to execute an activity or series of activities upon request” [PEM Specifications 3.1]. Types of services are: stand-alone service, web service, front-office services and back-office service, as specified by the has_type property.

Examples: The ARIADNE discovery service.

Maps to: AO_Service is equivalent to crmpe:PE8_E-Service which is a subclass of crm:E7_Activity

Domain of: is_accessible_at
has_functionality
has_consumed_media
has_produced_media
has_consumed_format
has_produced_format

has_supported_language
has_technical_support

4.3 AO_Object

Subclass of: AO_Entity

Superclass of:

Scope Note: This class includes as instances physical objects **of interest** to the ARIADNE infrastructure.

Examples: The walls of Troy

Maps to: AO_Object is a subclass of crm:E18_Physical_Thing

Domain of: has_time_interval
has_space_region
was_present_at

4.4 AO_Event

Subclass of: AO_Entity

Superclass of: AO_Activity

Scope Note: This class includes as instances events **of interest** to the ARIADNE RI, that is, events that play a role in the ARIADNE AC. An event is a “change of states in cultural, social or physical systems, regardless of scale, brought about by a series or group of coherent physical, cultural, technological or legal phenomena” [CIDOC CRM Specs 6.2].

Usage Note: It is the data curator who decides whether or not an event is of interest to ARIADNE and as such should or should not be modelled as an instance of this class. A typical criterion is whether or not the event is sufficiently well documented. If yes, then the event needs to be created to associate the resources with it. If not, the event remains “implicit”. For example, the making of a vase found in the context of an excavation and that has no data resources associated with it, is an event not of interest to ARIADNE. Consequently, it should not be created explicitly and the knowledge about it (e.g., who made the vase) should be modelled as direct properties of the vase (e.g., dc:creator). In contrast, the excavation itself might be of interest to ARIADNE if there are data resources associated with it; in this case, the excavation should be explicitly represented as an instance of this class.

Examples: The excavation of the walls of Troy for which the AC has data resources or the eruption of the Vesuvio that destroyed Pompei.

Maps to: AO_Event is a subclass of crm:E5_Event

Domain of: occurs_in
happens_during
contains_event

4.4.1 AO_Activity

Subclass of: AO_Event

Superclass of:

Scope Note: This class includes as instances activities **of interest** to the ARIADNE RI, that is, events that play a role in the ARIADNE AC. Activities are “actions intentionally carried out by actors that result in changes of state in the cultural, social, or physical systems documented” [CIDOC CRM Specs 6.2].

Usage Note: The same usage note as AO_Event applies.

Examples: The excavation of the walls of Troy for which the AC has data resources.

Maps to: AO_Event is a subclass of crm:E7_Activity

4.5 AO_Agent

Subclass of: AO_Entity

Superclass of: AO_Person, AO_Group

Scope Note: “This class comprises people, either individually or in groups, who have the potential to perform intentional actions of kinds for which someone may be held responsible.” [CIDOC CRM Specs 6.2].

Examples: The Principal Investigator or Director of an excavation event, the author of a scientific article.

Maps to: AO_Agent is a subclass of crm:E39_Actor

Domain of: has_name
has_agent_identifier
has_email
has_homepage

4.5.1 AO_Person

Subclass of: AO_Agent

Superclass of:

Scope Note: This class comprises individual human beings.

Examples: The Principal Investigator or Director of an excavation event, the author of a scientific article.

Maps to: AO_Person is a subclass of crm:E21_Person

Domain of: has_institution

4.5.2 AO_Group

Subclass of: AO_Agent

Superclass of:

Scope Note: This class comprises “gatherings or organizations that act collectively or in a similar way due to any form of unifying relationship” [CRM Spec 6.2].

Examples: The Archaeological Data Service

Maps to: AO_Group is a subclass of crm:E74_Group

4.6 AO_Temporal_Region

Subclass of: AO_Entity

Superclass of:

Scope Note: This class comprises temporal extents, having a beginning, an end and a non-zero duration. Temporal regions can have one of the following forms:

1. a temporal interval (e.g., from 155 BC to 243 AD). In this case the properties from and until are used to give the boundaries of the temporal interval
 2. any period identified by a name (e.g., “Neolithic”) expressed via the has_period or has_native_period property
- In the case where BC dates have to be supplied, a minus (-) sign could be used as indicated in the expanded Year representation of ISO 8601 (http://www.iso.org/iso/catalogue_detail?csnumber=40874).
 - In the case where reduced precision must be applied (e.g. where no day information is available) the respective part could be omitted (according to ISO 8601 reduced precision guidelines).

Examples: The interval in which the excavation of the walls of Troy has taken place

Maps to: AO_Temporal_Region is a subclass of crm:E52_Time-Span.

Domain of: has_period
has_native_period
from
until

4.7 AO_Spatial_Region

Subclass of: AO_Entity

Superclass of: AO_Spatial_Region_Point, AO_Spatial_Region_Polygon,
AO_Spatial_Region_BBox, AO_Spatial_Region_StdName

Scope Note: This class comprises spatial regions having one of the following forms:

1. a point on the surface of the Earth identified by latitude and longitude;
2. a polygon as represented by GIS systems;
3. a rectangular region on the surface of the Earth identified by its four vertices;
4. any region identified by a IRI in a standard gazetteer, such as Geonames for modern places, Pleiades for ancient places.

Each of these forms is modelled by a different sub-class of this class, as detailed in the rest of this Section. In addition, a spatial region may have a name, regardless of any other representation it may have. If the same region is represented in two different ways, then it will be an instance of the two corresponding classes.

Examples: The spatial region including Troy in the X sec. B.C.
Maps to: AO_Spatial_Region is a subclass of crm:E53_Place
Domain of: has_coordinate_system
has_place_name

4.7.1 AO_Spatial_Region_Point

Subclass of: AO_Spatial_Region
Superclass of:
Scope Note: This class comprises spatial regions given as points on the surface of the Earth identified by latitude and longitude.
Examples:
Maps to:
Domain of: has_latitude
has_longitude

4.7.2 AO_Spatial_Region_Polygon

Subclass of: AO_Spatial_Region
Superclass of:
Scope Note: This class comprises spatial regions given as a polygon as represented by GIS systems.
Examples:
Maps to:
Domain of: has_polygonal_representation

4.7.3 AO_Spatial_Region_BBox

Subclass of: AO_Spatial_Region
Superclass of:
Scope Note: This class comprises spatial regions given as rectangular regions on the surface of the Earth identified by its four vertices.
Examples:
Maps to:
Domain of: has_bounding_box_min_lat
has_bounding_box_min_lon
has_bounding_box_max_lat
has_bounding_box_max_long

4.7.4 AO_Spatial_Region_StdName

Subclass of: AO_Spatial_Region
Superclass of:
Scope Note: This class comprises spatial regions identified by a name expressed via the has_place_name property;
Examples:
Maps to:
Domain of: has_place_name

4.8 AO_Concept

Subclass of: AO_Entity

Superclass of:

Scope Note: This class comprises terms in thesauri, controlled vocabularies or any other reference resource providing concepts in a domain of interest. Since AO_Concept is equivalent to skos:Concept, all properties defined in SKOS can be used to represent the relevant aspects of a concept. For instance, the properties skos:broader and skos:narrower can be used to model concept taxonomies, while skos:broadMatch, skos:closeMatch, skos:exactMatch, skos:narrowMatch, skos:relatedMatch can be used to specify the kind of mapping leading to the instance of the concept, if any.

Examples: Any term in the Getty Art & Architecture Thesaurus

Maps to: AO_Concept is equivalent to crm:E55 Type and to skos:Concept

4.9 AO_Dimension

Subclass of: AO_Entity

Superclass of:

Scope Note: This class comprises values of quantifiable properties that can be determined by some measure, i.e. the radius of an approximated spatial region

Examples: a circle with a radius of 10 kilometers

Maps to: AO_Dimension is a sub-class of crm:E54 Dimension

4.10 AO_Digital_Media

Subclass of: AO_Individual_Data_Resource, PE18 Dataset

Superclass of: AO_Digital_Image

Scope Note: This class comprises digital media contributed to the ARIADNE Content cloud for the purpose of illustrating other resources.

Examples: (examples given in)

Maps to: AO_Digital_Media is a subclass of crmdig:D9_Data_Object (subclass of crm:E71_information_Object and E54_Dimension)

Domain of: is_mirror_of
is_visual_component_of
is_primary_visual_component_of
is_rendered_by

4.10.1 AO_Digital_Image

Subclass of: AO_Digital_Media

Superclass of:

Scope Note: This class comprises digital images contributed to the ARIADNE Content cloud for the purpose of illustrating other resources.

Examples: (examples given in [Section 2.1 Digital images](#))

Maps to: AO_Digital_Image is a subclass of crmdig:D9_Data_Object (subclass of crm:E71_information_Object and E54_Dimension)

Domain of: